

JEE MAIN REVISION MASTER: SOLID STATE 2026

0 QUESTIONS 0% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 23 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Solid State — Standard Calculations & Problem Archetypes	11 PYQs	Quantitative Analysis
High	Solid State — Applied Cases, Reactions & Traps	10 PYQs	Applied Topics
Moderate	Solid State — Core Principles & Governing Laws	3 PYQs	Core Principle

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Solid State — Standard Calculations & Problem Archetypes	$\rho = \frac{Z \cdot M}{a^3 \cdot N_A}$, $d = \frac{Z \cdot M}{a^3 \cdot N_A}$, $\rho = 10 \text{ g cm}^{-3}$, $N_A = 6 \times 10^{23}$
Solid State — Applied Cases, Reactions & Traps	$\alpha \neq \beta \neq \gamma \neq 90^\circ$, SrCl_2 , m^3
Solid State — Core Principles & Governing Laws	$[\text{PCl}_4]^+$, PCl_5

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Solid State — Standard Calculations & Problem Archetypes (High Yield — 11 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Solid State.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2023 Shift] Q#1 (MCQ): "A metal crystallises in a body centred cubic lattice with the metallic radius $\sqrt{3}a/4$. The volume of the unit cell in Å^3 ..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "If 'a' stands for the edge length of the cubic systems. The ratio of radii in simple cubic, body centred cubic and face centred cubic un..." [Ans: N/A]
- [2023 Shift] Q#4 (MCQ): "Match the column A (type of crystalline solid) with the column B (example for each type) A B P. Molecular solid i. SiC Q. Ionic solid ii. Mg..." [Ans: N/A]

3.2 Solid State — Applied Cases, Reactions & Traps (High Yield — 10 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Solid State.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Which of the following crystals has the unit cell such that $a=b \neq c$ and $\alpha=\beta=90^\circ$, $\gamma=120^\circ$?..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "The number of atoms in 4.5 g of a face-centred cubic crystal with edge length 300 pm is (Given : Density = 10 g cm^{-3} and $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$..." [Ans: N/A]
- [2022 Shift] Q#2 (MCQ): "Alkali halides do not show dislocation defect because..." [Ans: N/A]

3.3 Solid State — Core Principles & Governing Laws (Moderate Yield — 3 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Solid State.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Solid State.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Solid State PYQs.

Representative Source PYQs:

- [2023 Shift] Q#2 (MCQ): "In solid state, PCl_5 is a/an..." [Ans: N/A]
- [2020 Shift] Q#3 (MCQ): "Silicon doped with gallium forms..." [Ans: N/A]
- [2018 Shift] Q#3 (MCQ): "In FCC, the unit cell is shared equally by how many unit cells?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#2)

The number of atoms in 4.5 g of a face-centred cubic crystal with edge length 300 pm is (Given : Density = 10 g cm^{-3} and $N_A = 6.022 \times 10^{23}$)

- (A) 6.6×10^{20}
- (B) 6.6×10^{23}
- (C) 6.6×10^{19}
- (D) 6.6×10^{22}

Correct Answer: None

Solution Analysis: Given, Face centred crystal, so $Z=4$ edge length $(a)=300 \text{ pm}=300 \times 10^{-10} \text{ cm}$ density $(d)=10 \text{ g cm}^{-3}$ $N_A=6.022 \times 10^{23}$ Using the formula $d = \frac{Z \times M}{N_A \times a^3}$ $M = \frac{d \times N_A \times a^3}{Z}$ $= \frac{10 \times 6.022 \times 10^{23} \times (300 \times 10^{-10})^3}{4} \Rightarrow M = 40.65 \text{ g}$ So, 40.65 g contains 6.022×10^{23} atoms Let 4.5 g contains x atom $\Rightarrow \frac{6.022 \times 10^{23}}{40.65} \times 4.5 = x \Rightarrow x = 6.6 \times 10^{22}$

Benchmark Problem 2: KCET 2023 (Q#1)

A metal crystallises in a body centred cubic lattice with the metallic radius $\frac{\sqrt{3}}{4}a$. The volume of the unit cell in m^3 is

- (A) 64×10^{-29}
- (B) 4×10^{-29}
- (C) 6.4×10^{-29}
- (D) 4×10^{-10}

Correct Answer: None

Solution Analysis: $r = \frac{\sqrt{3}}{4}a$ $a = \frac{4r}{\sqrt{3}} = \frac{4 \times \frac{\sqrt{3}}{4} \times 10^{-10}}{\sqrt{3}} = 10^{-10} \text{ m}$ $a^3 = 10^{-30} \text{ m}^3$ Thus, the volume of the unit cell is 10^{-30} m^3 .

Benchmark Problem 3: KCET 2023 (Q#2)

In solid state, PCl_5 is a/an

- (A) octahedral structure
- (B) ionic solid with $[\text{PCl}_6]^+$ and $[\text{PCl}_4]^-$
- (C) ionic solid with $[\text{PCl}_4]^+$ and $[\text{PCl}_6]^-$
- (D) covalent solid present in the form of P_2Cl_{10}

Correct Answer: None

Solution Analysis: PCl_5 is an ionic solid composed of $[\text{PCl}_4]^+$ ions and $[\text{PCl}_6]^-$ ions. In an ionic solid, there is a transfer of electrons between atoms, resulting in the formation of these cations and anions. These opposite charged ions are held together by electrostatic forces forming a crystal lattice structure.